

CourseCode:B23EC4101					
SAGIRAMAKRISHNAMRAJUENGINEERINGCOLLEGE(A)					R23
IV B.Tech. I Semester MODE LQUESTIONPAPER					
CELLULAR & MOBILE COMMUNICATIONS					
For ECE					
Time:3Hrs.			Max.Marks:70M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 X 2=20 Marks					
			CO	KL	M
1	a).	What are the limitations of conventional wireless systems?	1	2	2
	b).	Define frequency reuse ratio.	1	3	2
	c).	Define co-channel interference (CCI).	2	2	2
	d).	What happens when transmit power is reduced?	2	3	2
	e).	Define channel borrowing.	3	2	2
	f).	What is antenna height gain?	3	2	2
	g).	Write a short note about Intrasytem handoffs	4	2	2
	h).	Discuss about handoff initiation	4	3	2
	i).	What are GSM channels?	5	2	2
	j).	What is the main advantage of CDMA over FDMA/TDMA?	5	2	2
5x 10 = 50 Marks					
UNIT-I					
2.	a).	Explain about consideration of the components of Cellular system	1	3	5
	b).	Compare 1G, 2G, 3G, and 4G cellular systems in terms of technology, services, and data rates.	1	3	5
OR					
3.	a).	Derive the co-channel interference reduction factor for seven cell reuse pattern	1	2	5
	b).	Discuss about Trucking and GOS.	1	3	5
UNIT-II					
4.	a).	Explain the measurement of real time Co-Channel interference.	2	3	5
	b).	With neat sketch explain the effect of Non-co channel interference-adjacent channel interference.	2	3	5
OR					
5.	a).	Explain co-channel interference (CCI) and derive the expression for Signal-to-Interference Ratio (C/I)..	2	3	5
	b).	Discuss the near-end far-end interference problem with suitable examples.	2	2	5
UNIT-III					

6.	a).	What is the importance of frequency management chart?	3	3	5
	b).	Discuss about fixed channel and non-fixed channel assignment	3	2	5
		OR			
7.	a).	Explain signal reflections in flat and hilly terrain and their impact on communication.	3	3	5
	b).	Explain in detail about near and long-distance mobile propagation.	3	3	5
		UNIT-IV			
8.	a).	How interference can be reduced by using the directional antennas at cell site.	4	3	5
	b).	Describe the effects of cell site antenna heights and signal coverage cells.	4	3	5
		OR			
9.	a).	Write about forced handoff and delayed handoff mechanisms in detail.	4	2	5
	b).	Compare soft handoff and hard handoff in detail.	4	3	5
		UNIT-V			
10.	a).	Explain the GSM architecture with a neat diagram, including BSS, NSS, and OSS.	5	3	5
	b).	Compare FDMA, TDMA, and CDMA in terms of capacity, efficiency, and complexity.	5	2	5
		OR			
11.	a).	Explain the concept of MIMO systems and their advantages in wireless communication.	5	2	5
	b).	Discuss the technological advancements in 5G compared to 4G	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

Estd. 1980

AUTONOMOUS

Course Code: B23HS4104					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
MANAGEMENT SCIENCE					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What is time study?	1	1	2
	b).	Define organization.	1	1	2
	c).	Define Plant Layout.	2	3	2
	d).	What is a travel chart?	2	3	2
	e).	Define critical path.	3	2	2
	f).	What is process control?	3	2	2
	g).	What is inventory?	4	2	2
	h).	Classify A, B, and C items.	4	2	2
	i).	Define Job Evaluation.	5	1	2
	j).	What is Marketing Mix?	5	1	2
5 x 10 = 50 Marks					
		UNIT-I			
2.	a).	Define management? Explain the functions of management.	1	2	5
	b).	Explain Maslow's hierarchy of human needs?	1	2	5
		OR			
3.	a).	Discuss and differentiate Taylor's and Fayol's approaches to management.	1	2	5
	b).	Compare Line, Functional, and Matrix Organizations.	1	2	5
		UNIT-II			
4.	a).	Explain the concept of plant location and discuss the factors affecting plant location.	2	3	5
	b).	Explain different types of plant layouts with suitable examples.	2	2	5
		OR			
5.	a).	Define work study and explain its objectives and importance.	2	3	5
	b).	Explain various charts used in Method Study.	2	2	5
		UNIT-III			

6.	a).	Explain Project Management and Network Modeling.	3	2	5
	b).	Define Critical Path and explain its significance.	3	2	5
		OR			
7.	a).	Explain variables and attributes in quality control with suitable examples	3	2	5
	b).	Explain the role of statistical techniques in quality control.	3	2	5
		UNIT-IV			
8.	a).	Explain various stores records maintained in an organization.	4	3	5
	b).	Discuss the duties and responsibilities of a purchase manager.	4	2	5
		OR			
9.	a).	Discuss the advantages and limitations of ABC analysis.	4	3	5
	b).	Explain various inventory control methods.	4	2	5
		UNIT-V			
10.	a).	Explain the importance and objectives of HRM.	5	3	5
	b).	Describe different methods of Job Evaluation.	5	3	5
		OR			
11.	a).	Define Marketing and explain its functions.	5	3	5
	b).	Explain strategies in different stages of Product Life Cycle.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

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Estd. 1980

AUTONOMOUS

Course Code:B23EC4102					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
ANALOG VLSI LAYOUT TECHNIQUES					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What is threshold voltage (V_t) in a MOSFET?	1	1	2
	b).	What is the purpose of design rules in VLSI layout?	1	2	2
	c).	What is noise in a MOSFET?	2	1	2
	d).	Define fingers in MOS transistor layout.	2	1	2
	e).	Define shielding in VLSI layout.	3	1	2
	f).	What is the antenna effect in VLSI?	3	1	2
	g).	What is the advantage of a cascode structure?	4	2	2
	h).	What causes Well Proximity Effect?	4	2	2
	i).	Mention one advantage of FinFET over planar MOSFET.	5	2	2
	j).	What is interdigitated layout?	5	1	2
5 x 10 = 50 Marks					
UNIT-1					
2.		Explain the p-well process for a CMOS inverter with neat diagrams.	1	2	10
OR					
3.	a).	Derive the expression for the ratio between $Z_{p.u}$ and $Z_{p.d}$ if an NMOS inverter is to be driven from another NMOS inverter	1	3	5
	b).	Explain different scaling factors for device parameters?	1	2	5
UNIT-2					
4.	a).	Explain common centroid layout technique with neat diagram and advantages.	2	2	5
	b).	Describe fingers and multipliers in MOS layout and their advantages.	2	2	5
OR					
5.	a).	Write short notes on interdigitized layout and where it is used.	2	1	5
	b).	Explain device sizing in analog layout and its impact on performance.	2	2	5
UNIT-3					
6.	a).	Explain electromigration, its causes and effects on IC reliability.	3	2	5

	b).	Explain shielding techniques used to reduce noise and interference.	3	2	5
		OR			
7.	a).	Explain latch-up in CMOS, its mechanism, effects, and prevention techniques.	3	2	5
	b).	Discuss coupling in VLSI layout and its effect on signal integrity.	3	2	5
		UNIT-4			
8.	a).	Discuss the importance of isolation techniques in analog layout design.	4	2	5
	b).	Explain Shallow Trench Isolation (STI) with structure and advantages.	4	2	5
		OR			
9.	a).	Describe the structure and function of deep N-well, and how it improves isolation.	4	2	5
	b).	Discuss the Well Proximity Effect (WPE), its causes and impact on device performance.	4	2	5
		UNIT-5			
10.	a).	Explain fabrication challenges in FinFET technology.	5	2	5
	b).	Explain op-amp layout using common centroid matching and its importance.	5	2	5
		OR			
11.	a).	Explain FinFET technology, its structure and working principle.	5	2	5
	b).	List the differences between common centroid and interdigitated matching techniques.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

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Course Code: B23EC4103					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
EMBEDDED COMMUNICATION PROTOCOLS & INTERFACES					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Explain half duplex and full duplex communication?	1	2	2
	b).	Discuss about RS485 communication?	1	2	2
	c).	Illustrate Fire wire standard?	2	2	2
	d).	Write a short note on the PCI bus?	2	1	2
	e).	Explain Transport Control Protocol?	3	1	2
	f).	Discuss about IEEE 802.3 standard?	3	2	2
	g).	What are the applications of Wireless Sensor Networks in network embedded systems?	4	2	2
	h).	What are the design Challenges observed in network embedded systems?	4	1	2
	i).	list out the various MAC protocols?	5	2	2
	j).	Explain about CDMA?	5	1	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Compare synchronous and asynchronous communications	1	2	5
	b).	Explain HDLC protocol?	1	2	5
		OR			
3.	a).	Explain serial data communication using SPI and SCI ports.	1	2	5
	b).	Illustrate about UART protocol?	1	2	5
		UNIT-2			
4.		Explain in detail about CAN Bus & USB Bus	2	2	10
		OR			
5.		Discuss about parallel bus device protocols?	2	2	10
		UNIT-3			
6.	a).	Discuss about HTTP and internet protocol ?	3	2	5

	b).	explain about Bus and star topologies	3	2	5
		OR			
7.	a).	Explain CSMA/CA protocol?	3	2	5
	b).	Write short note on Hubs, switches, and Bridges	3	2	5
		UNIT-4			
8.		Discuss about Networked Embedded Systems in Building Automation	4	2	10
		OR			
9.		Discuss about Networks Embedded Systems in Industrial Automation	4	2	10
		UNIT-5			
10.		Compare CDMA & FDMA?	5	2	10
		OR			
11.	a).	Write a note on IEEE 802.11 standard?	5	2	5
	b).	Discuss about the ZIGBEE protocol?	5	2	5

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KL-KNOWLEDGE LEVEL

M-MARKS

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Course Code: B23EC4104					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
CODING THEORY AND APPLICATIONS					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What is joint entropy?	1	1	2
	b).	What is the significance of conditional entropy?	1	2	2
	c).	What is mutual information?	2	1	2
	d).	Differentiate between noiseless and noisy channels.	2	2	2
	e).	What is syndrome in linear block codes?	3	1	2
	f).	Differentiate between error detection and error correction.	3	2	2
	g).	What is the algebraic structure of cyclic codes?	4	1	2
	h).	What is the purpose of ARQ in communication systems	4	2	2
	i).	Define code rate in convolution coding.	5	1	2
	j).	Why are turbo codes used in modern communication systems?	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Discuss the steps involved in Shanon-Fano algorithm	1	2	5
	b).	An analog signal is band limited to 800 Hz, sampled at the Nyquist rate, and the samples are quantized into four levels. The quantization levels are assumed independent and occur with probabilities (1/8, 1/8, 3/8, and 3/8). Find the entropy H(X) and information rate R of the source.	1	3	5
		OR			
3.	a).	Discuss in detail about Shannon source coding theorem.	1	2	5
	b).	A source emits messages with probabilities (1/2, 1/4, 1/8, 1/16, 1/32, and 1/32). Calculate (i) entropy of the source, (ii) Apply Shannon-Fano algorithm to devise a binary code for this source and find its coding efficiency and redundancy	1	3	5
		UNIT-2			
4.	a).	Derive the expression for the channel capacity of AWGN channel.	2	3	5
	b).	State & explain the Shannon's noisy channel coding theorem. Find the	2	3	5

		channel capacity of a binary symmetric channel.			
		OR			
5.	a).	Derive the expressions for mutual entropy, mutual information and joint probabilities	2	3	5
	b).	Find the mutual entropy $H(X;Y)$ for the suitable channel. Consider suitable values for Joint probabilities $P(X;Y)$. Assume source symbols are equally likely	2	3	5
		UNIT-3			
6.	a).	Discuss about Linear block Codes.	3	2	4
	b).	Demonstrate that (7,4) Hamming code can correct a single error & can detect a double error by syndrome decoding.	3	3	6
		OR			
7.	a).	Write about single parity check bit codes.	3	2	4
	b).	Explain the generation of a linear systematic (n, k) blocks code using generator matrix. Define minimum Hamming distance d_{min} of a code. What is the relation between d_{min} and the error correcting capacity of a code?	3	3	6
		UNIT-4			
8.	a).	Write about BCH codes.	4	2	4
	b).	Find the generator polynomial & parity check polynomial for a linear (7,4) systematic cyclic code. Use them to code and decode a message 1010.	4	3	6
		OR			
9.	a).	Discuss briefly about ARQ's.	4	2	5
	b).	Explain about generation of cyclic codes using generator matrix	4	2	5
		UNIT-5			
10.	a).	Write in detail about structural properties of convolution codes.	5	2	4
	b).	Draw the structure of a rate 1/2 Convolutional coder for $g_1 = [1\ 0\ 1]$ and $g_2 = [0\ 1\ 1]$. Explain the encoding process. Construct the state diagram, trellis diagram & code tree. Find the coder output for input data = [1 0 1 0 1].	5	3	6
		OR			
11.	a).	Discuss about Burst error correction.	5	2	5
	b).	Write a short note on turbo encoder.	5	2	5

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KL-KNOWLEDGE LEVEL

M-MARKS

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Course Code: B23EC4105					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
OPTICAL FIBER COMMUNICATION					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Illustrate Critical Angle of a Optical Fiber?	1	2	2
	b).	What is V-number in connection with optical fibers?	1	3	2
	c).	Define attenuation in fibers.	2	2	2
	d).	State different modes in fibers.	2	1	2
	e).	What is the need of fiber connectors and mention its applicable areas.	3	1	2
	f).	List out fiber fabrication methods.	3	2	2
	g).	Draw the radiation patterns for surface emitting and edge emitting LED's.	4	2	2
	h).	Define responsivity of a photo detector.	4	1	2
	i).	What are the various noise sources in an optical receiver?	5	2	2
	j).	What is meant by FDDI.	5	1	2
5 x 10 = 50 Marks					
UNIT-1					
2.	a).	Derive Numerical Aperture.	1	2	5
	b).	A multimode step index fiber with a core diameter of 80 μm and a relative index difference of 1.5% is operating at a wave of 0.85 μm if the core refractive index 1.48, Calculate (i) the normalized frequency for the fiber (ii) the number of guided modes?	1	3	5
OR					
3.	a).	Discuss in detail about advantages of optical fiber communications?	1	2	5
	b).	Calculate Numerical Aperture of a silica optical fiber with a core diameter large enough to be considered by ray theory analysis has a core refractive index of 1.50 and a cladding refractive index of 1.47?	1	3	5
UNIT-2					
4.	a).	Discuss about pulse broadening in graded index fibers.	2	2	5
	b).	Explain Material dispersion in fiber.	2	3	5
OR					
5.	a).	Discuss about fiber bending loss.	2	2	5
	b).	Explain Intermodal dispersion in fiber.	2	3	5

		UNIT-3			
6.	a).	Explain the splicing procedure and compare any two splicing techniques with a neat sketch.	3	2	5
	b).	Discuss about fiber alignment and joint loss in optical fiber communication.	3	2	5
		OR			
7.	a).	Describe the basic structure of the cable and explain all the structural elements.	3	3	5
	b).	Discuss about fiber connectors with neat sketches	3	2	5
		UNIT-4			
8.	a).	Describe the two basic LED structures that are suitable for optical fibers with neat schematics.	4	3	5
	b).	A double heterojunction INGaAsP LED emitting at a peak wavelength of 1310nm has radiative and nonradiative recombination times 25ns and 90ns, respectively. The drive current is 35 mA. Calculate the internal quantum efficiency and the internal power level.	4	2	5
		OR			
9.	a).	List and explain all the injection laser characteristics	4	3	5
	b).	Compare and draw a conclusion for two different semiconductor photodiodes with and without internal gain.	4	4	5
		UNIT-5			
10.	a).	Design a single mode fiber link by calculating the length of the fiber with an attenuation of 0.3dB/km and with the required power margin 3dB. The fiber link includes 5 splices at 0.02dB/Splice, 4 connectors at 0.2dB/Connector, transmitter power of -10 dBm and the receiver sensitivity of -25 dBm.(Use graphical method)	5	4	5
	b).	Explain the operation of a fundamental Fiber optic receiver	5	4	5
		OR			
11.	a).	Explain point to point links with relevant diagrams.	5	3	5
	b).	Describe the operation of the WDM system with a neat diagram.	5	4	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

Course Code: B23EC4106					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
DSP PROCESSORS AND ARCHITECTURES					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Differentiate between discrete-time and continuous-time signals	1	2	2
	b).	Define a linear time-invariant (LTI) system	1	1	2
	c).	Define Address Generation Unit (AGU).	2	1	2
	d).	Explain programmability in DSP processors.	2	2	2
	e).	Explain data addressing modes of TMS320C54XX.	3	2	2
	f).	Describe memory space organization in TMS320C54XX.	3	2	2
	g).	Define Blackfin processor	4	1	2
	h).	Describe shifter instructions in DSP processors	4	2	2
	i).	Explain the advantages of DMA.	5	2	2
	j).	Explain interrupt-driven I/O	5	2	2
5 x 10 =50Marks					
UNIT-1					
2.	a).	Explain the basic Digital signal processing system with the help of block diagram.	1	2	5
	b).	Explain in detail about sources of errors in DSP implementations.	1	2	5
OR					
3.	a).	Discuss about the DFT and FFT calculations in Digital signal processors.	1	2	5
	b).	Write about fixed point and floating-point formats	1	2	5
UNIT-2					
4.	a).	Discuss in brief about the data addressing capabilities of programmable DSP devices with examples.	2	3	5
	b).	Explain the operation of address generation unit in a DSP processor with neat block diagram.	2	3	5
OR					
5.	a).	What are the features for external interfacing in a DSP processor?	2	2	5
	b).	Write about the programmability and program execution sequence of a DSP	2	2	5

		UNIT-3			
6.	a).	What is the need an interrupt in a processor? Write about Interrupts of TMS320C54XX Processor.	3	3	5
	b).	Explain the concept of Pipelining for speeding up the execution of an Instruction.	3	3	5
		OR			
7.	a).	Write about different on chip peripherals of TMS320C54XX Processor.	3	3	5
	b).	Explain about the memory space organization of TMS320C54XX Processor.	3	3	5
		UNIT-4			
8.	a).	Explain the micro signal architecture in detail.	4	3	5
	b).	Discuss in brief about the basic peripherals in an a log device family of DSP devices	4	2	5
		OR			
9.	a).	Explain the architecture of Black fin processor with neat circuit diagram	4	2	5
	b).	Explain the architecture of ADSP 2100	4	2	5
		UNIT-5			
10.	a).	Draw and explain the block diagram of memory interface for TMS320C5416 processor.	4	3	5
	b).	How does DMA help in increasing the processing speed of a DSP processor?	4	2	5
		OR			
11.	a).	How an external bus can be interfaced with TMS320C54XX Processor? Explain	5	4	5
	b).	Explain the Parallel I/O and Programmable I/O	5	4	5

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Course Code: B23EC4107					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
DESIGN FOR TESTABILITY					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Distinguish between defect, error, and fault	1	2	2
	b).	List different levels of fault models	1	1	2
	c).	State the need for simulation in design verification	2	2	2
	d).	Compare compiled code and event-driven simulation.	2	2	2
	e).	What is partial scan design?	3	2	2
	f).	Define scan design	3	1	2
	g).	Differentiate Test-Per-Clock and Test-Per-Scan BIST	4	2	2
	h).	State advantages of BIST	4	2	2
	i).	Describe pin description in BSDI	5	2	2
	j).	List the ports in boundary scan architecture	5	1	2
5 x 10 =50Marks					
UNIT-1					
2.	a).	Describe Digital and Analog VLSI testing.	1	2	5
	b).	Describe Functional Versus Structural Testing	1	2	5
OR					
3.		Explain how Stuck-at Fault can be used to cover faults in a system.	1	2	10
UNIT-2					
4.		Discuss about modelling Circuits for Simulation	2	2	10
OR					
5.		Explain the terms a) Compiled code Simulation. b) Event Driven Simulation	2	3	10
UNIT-3					
6.		With neat diagrams, explain partial-scan design methods.	3	3	5
OR					
7.		Explain combinational SCOAP measures with a circuit example.	3	3	5

		UNIT-4			
8.		Explain about BIST process with necessary diagrams.	4	3	5
		OR			
9.		Explain Test-Per Scan BIST System using D flip-flop mode and LFSR mode.	4	3	10
		UNIT-5			
10.		Describe about boundary scan standard instructions.	5	2	10
		OR			
11.		Explain about Boundary Scan Description Language components and pin description.	5	3	10

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SRKR
ENGINEERING COLLEGE
AUTONOMOUS

Course Code: B23EC4108					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
PHYSICAL DESIGN					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define VLSI Physical Design.	1	1	2
	b).	What is the role of automation in VLSI?	1	2	2
	c).	List the objectives of floor planning.	2	1	2
	d).	Why should macros be placed near I/O pins?	2	2	2
	e).	Define congestion analysis	3	2	2
	f).	What is timing analysis in placement?	3	2	2
	g).	Mention any two challenges in routing.	4	1	2
	h).	What is performance-driven design flow?	4	2	2
	i).	Define Static Timing Analysis	5	1	2
	j).	Why are hold checks required?	5	2	2
5 x 10 =50Marks					
UNIT-1					
2.	a).	Explain VLSI Physical Design and its objectives in detail.	1	3	5
	b).	Analyze the different stages involved in ASIC design flow.	1	4	5
OR					
3.	a).	Explain the basic inputs required for block-level physical design implementation.	1	2	5
	b).	Discuss the significance of LEF and DEF files in physical design.	1	2	5
UNIT-2					
4.		Explain full chip floorplanning and block-level floorplanning with neat diagrams.	2	2	10
OR					
5.		Explain static and dynamic power consumption with equations and examples	2	2	10
UNIT-3					
6.	a).	Explain the goals and objectives of placement in VLSI physical design.	3	2	5
	b).	Analyze the effects of clock skew and latency on circuit performance.	3	4	5

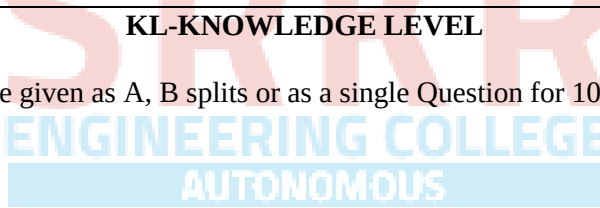
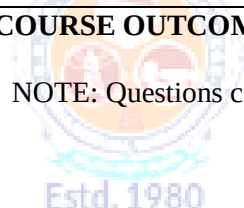
		OR			
7.	a).	Explain congestion analysis methods and their importance in placement.	3	2	5
	b).	Analyze challenges in CTS and techniques to minimize skew and power.	3	4	5
		UNIT-4			
8.	a).	Explain global routing and detailed routing with suitable examples.	4	2	5
	b).	Analyze the impact of parasitic elements on circuit performance.	4	4	5
		OR			
9.	a).	Discuss the objectives and challenges of routing in VLSI design.	4	2	5
	b).	Analyze trade-offs involved in timing optimization (area, power, delay).	4	4	5
		UNIT-5			
10.		Explain setup time and hold time checks with timing diagrams	5	2	10
		OR			
11.	a).	Discuss the importance of timing analysis in VLSI design.	5	2	5
	b).	Analyze the impact of parasitic elements on circuit performance	5	4	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks



Course Code: B23EC4109					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
INTERNET OF THINGS & ITS APPLICATIONS					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Explain the fundamental difference between M2M Communication and IoT.	1	2	2
	b).	Describe the primary purpose of the "Network Layer".	1	2	2
	c).	Define the operational frequency bands.	2	2	2
	d).	Define the role of a Broker in an MQTT-based IoT architecture.	2	1	2
	e).	Identify the primary difference between a Microcontroller (MCU) and a General-Purpose Microprocessor (MPU).	3	1	2
	f).	Explain the function of an "IoT Platform" in managing device communication.	3	2	2
	g).	Define "Data Analytics" in the context of IoT.	4	1	2
	h).	Explain the fundamental difference between "Data in Motion" and "Data at Rest".	4	2	2
	i).	List four key IoT electronic equipments used for environmental monitoring in "Smart Buildings."	5	1	2
	j).	Define "Industry 4.0".	5	1	2
5 x 10 =50Marks					
		UNIT-1			
2.	a).	Detail the functional blocks of an IoT ecosystem.	1	3	5
	b).	Compare the IoTWF Standardized Architecture with the Simplified IoT Architecture .	1	3	5
		OR			
3.	a).	Explain the IoT World Forum (IoTWF) reference model.	1	2	5
	b).	Evaluate the Core IoT Functional Stack and discuss the complexities of data management.	1	4	5
		UNIT-2			
4.		Explain the architecture of LoRaWAN, focusing on the roles of End Devices, Gateways, and the Network Server in ensuring long-range connectivity.	2	2	10

		OR			
5.		Detail the operational principles of the Constrained Application Protocol (CoAP).	2	3	10
		UNIT-3			
6.		Describe the step-by-step IoT Design Methodology, from initial requirements gathering to the final deployment and integration phase.	3	2	10
		OR			
7.		Detail the IoT System Building Blocks and explain how the "Communication Block" interfaces with the "Services Block" in a cloud-based architecture.	3	3	10
		UNIT-4			
8.		Evaluate the benefits and trade-offs of Computing Using a Cloud Platform for large-scale M2M services versus using a decentralized Fog architecture.	4	4	10
		OR			
9.		Explain the complete Data Lifecycle in IoT, from initial acquisition and organization to the generation of actionable insights.	4	2	10
		UNIT-5			
10.		Discuss the implementation of IoT in Building Management Systems (BMS) to optimize HVAC and lighting for sustainability.	5	2	10
		OR			
11.		Discuss the integration of IoT electronic equipments in Smart Cities to manage traffic congestion and public safety infrastructures.	5	3	10

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

Course Code: B23EC4110					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
DIGITAL IMAGE PROCESSING					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Discuss about photo receptors located on retina?	1	1	2
	b).	Write short note on 2D-DFT and its properties	1	1	2
	c).	Define Histogram of an image?	2	1	2
	d).	What is the objective of sharpening spatial filter?	2	2	2
	e).	In an image compression model, how irrelevant information was removed?	3	1	2
	f).	Discuss about variable length coding?	3	2	2
	g).	Discuss about image gradient?	4	2	2
	h).	Outline the image segmentation based on image histogram?	4	1	2
	i).	Differentiate Hue and Saturation?	5	2	2
	j).	Which color model is well suited for color description?	5	1	2
5 x 10 =50Marks					
		UNIT-1			
2.	a).	How do you represent a digital image using sampling and quantization?	1	2	5
	b).	Interpret the basic relationships between pixels in a digital image?	1	2	5
		OR			
3.	a).	Outline the fundamental steps used in digital image processing?	1	2	5
	b).	Discuss the need of image transforms and Explain about DCT and its properties	1	2	5
		UNIT-2			
4.		Explain the concept of Histogram Equalization technique for Image enhancement with an example	2	3	10
		OR			
5.	a)	Discuss about adaptive median filtering?	2	2	5
	b)	Discuss briefly about important noise probability functions with neat plots?	2	2	5
		UNIT-3			

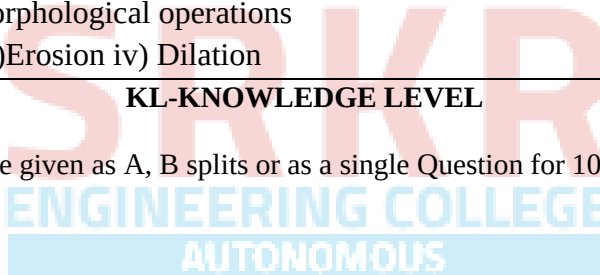
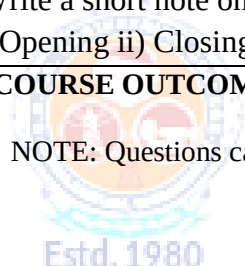
6.	a).	Discuss about image redundancies and outline the functioning of a general image compression system with a neat sketch?	3	2	5
	b).	Explain the JPEG standard used in image compression.	3	2	5
		OR			
7.		Illustrate Huffman coding process with an example	3	2	10
		UNIT-4			
8.	a).	Differentiate edge-based segmentation and region-based segmentation?	4	2	5
	b).	Illustrate about basic global thresholding?	4	2	5
		OR			
9.	a).	Explain about Region Splitting and Merging with an example.	4	2	5
	b).	How do you detect point, line and edges in an image during image segmentation?	4	2	5
		UNIT-5			
10.		Discuss about different color models used in image representation?	5	2	10
		OR			
11.	a).	Explain how to convert RGB to HSI color model?	5	3	5
	b).	Write a short note on morphological operations i) Opening ii) Closing iii) Erosion iv) Dilation	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks



Course Code:B23EC4111					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
SATELLITE COMMUNICATIONS					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Explain the significance of orbital perturbations in satellite communication.	1	3	2
	b).	Describe the impact of orbital effects on communication system performance.	1	4	2
	c).	Explain the role of the Attitude and Orbit Control System (AOCS) in satellite operations.	2	3	2
	d).	Analyze the importance of space qualification for satellite equipment reliability.	2	4	2
	e).	Calculate the system noise temperature and explain its impact on satellite link performance.	3	3	2
	f).	Analyze the significance of the G/T ratio in satellite communication link design.	3	4	2
	g).	Compare FDMA and TDMA in terms of bandwidth utilization and efficiency.	4	3	2
	h).	Analyze the advantages of CDMA over other multiple access techniques in satellite communication.	4	4	2
	i).	Explain the working principle of the NAVSTAR GPS system.	5	3	2
	j).	Analyze the differences between GPS and other global navigation systems.	5	4	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Express the three Kepler's laws of planetary motion.	1	2	5
	b).	Explain about various orbital parameters used in orbital analysis of a satellite.	1	3	5
		OR			
3.	a).	Derive the suitable equations for look angles and the range for Geostationary Satellite.	1	3	6
	b).	Explain about different launch vehicles.	1	2	4
		UNIT-2			

4.	a).	Examine how the attitude and orbit control system (AOCS) is Achieved through spin stabilization systems? Give necessary diagrams.	2	3	5
	b).	Explain the reason for inter modulation noise originating in a satellite link.	2	2	5
		OR			
5.	a).	Explain TT and C system in detail.	2	3	5
	b).	Explain about various power sources for the satellite.	2	2	5
		UNIT-3			
6.	a).	Calculate the carrier-to-noise ratio for the combined uplink and Down link communication.	3	3	5
	b).	State how inter modulation noise originates in a satellite link and Describe how it is reduced?	3	3	5
		OR			
7.	a).	Illustrate in detail about the free space transmission	3	2	5
	b).	Summarize the sources of noise in satellite communication. What is the importance of noise temperature in link design?	3	3	5
		UNIT-4			
8.	a).	Illustrate the features of various multiple access schemes deployed for satellite access and compare it.	4	3	5
	b).	Determine the limitations of CDMA.	4	2	5
		OR			
9.	a).	Explain direct sequence spread spectrum communication in detail.	4	2	5
	b).	Compare single access and multiple accesses.	4	3	5
		UNIT-5			
10.	a).	State and explain the working principle behind GPS?	5	2	5
	b).	Explain Evolution of GPS with neat diagrams.	5	3	5
		OR			
11.	a).	Explain the function of different segments in the operation of GPS in detail with necessary diagrams.	5	2	6
	b).	Compare GPS, GLONASS and GALILEO in terms of orbit configuration and services.	5	3	4

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

Course Code: B23EC4112					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
LOW POWER VLSI DESIGN					
For ECE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Describe the structure of an MIS capacitor.	1	2	2
	b).	Explain velocity saturation phenomenon.	1	2	2
	c).	Explain subthreshold leakage current.	2	2	2
	d).	Describe glitching power dissipation.	2	2	2
	e).	Explain deep submicron design issues.	3	2	2
	f).	Explain transistor stacking method	3	2	2
	g).	Describe clocked design style	4	2	2
	h).	Define domino logic	4	1	2
	i).	Explain pass transistor logic	5	2	2
	j).	Compare NMOS and CMOS logic styles.	5	2	2
5 x 10 =50Marks					
UNIT-1					
2.	a).	What is threshold voltage of MOSFET? Write down the mathematical expression. What is the effect of body voltage on threshold voltage?	1	2	5
	b).	What is Hot electron effect? Explain with diagram	1	2	5
OR					
3.		Draw and explain the energy band diagrams of MIS structure at different bias conditions.	1	2	10
UNIT-2					
4.		Explain the transistor leakage mechanisms of deep submicron transistors	2	2	10
OR					
5.		Explain about Short circuit power dissipation, glitching power dissipation	2	2	10
UNIT-3					
6.		Explain about Self Reverse Bias. Explain about Stacking effect in 2-input NAND gate.	3	2	5

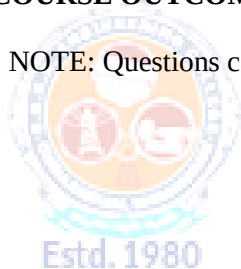
		OR			
7.		What are short channel effects? How to minimize short channel effects?	3	2	5
		UNIT-4			
8.	a).	What are the advantages of Domino Circuits	4	2	5
	b).	Explain about Domino logic. Explain the operation of domino NAND gate	4	2	5
		OR			
9.		Discuss in detail about Differential Current Switch Logic	4	2	10
		UNIT-5			
10.		Implement the function $F=A'B$ using 1.Fully complementary logic 2. Pass transistor Logic 3.pseudo NMOS Logic	5	3	10
		OR			
11.		Discuss in detail about differential cascade voltage switch logic (DCVS)	5	2	10

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

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